Back To FORM @ TEMPLATE

1	Course Name:	Fundamental of Artificial Intelligence
	Course Code:	PROG2053
	Course Classification:	Elective (core)
2	Synopsis:	The Artificial Intelligence (AI) course provides students with a comprehensive understanding of the field, covering foundational concepts, advanced techniques implemented in AI , and also the real-world applications.
3	Name(s) of Academic Staff:	1 Noor Fatihah Mazlam
		2
		3
4	Semester and Year offered:	Year Offered 2 Semester 2 *Remarks:*
5	Credit Value:	3 3
6	Pre-requisite/ co-requisite (if any):	none

7

Course Learning Outcomes (CLO)		
	CLO1	To gain a robust comprehension of the essential concepts, principles, and methodologies related to Artificial Intelligence. (C1,PLO1)
	CLO2	To demonstrate knowledge and understanding of the principal achievement, shortcomings, and various methods of artificial intelligence. (C3, PLO2)
	CLO3	To understand the ethical implications and societal impacts of AI technologies, considering biases, privacy, and responsible AI development. (C2, PLO11)

8 Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment Methods

Course Learning Outcomes	Programme Learning Outcomes (PLO)												Teaching Methods	Assessment Methods
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11			
CLO1	√												Lecture	st, Final Examination, Assignment, Proje
CLO2		√											Lecture and case study	st, Final Examination, Assignment, Proje
CLO3											√		Lecture and case study	st, Final Examination, Assignment, Proje
Mapping with MQF Cluster of Learning Outcomes	C1													
		C2												
											C5			

Indicate the primary causal link between the CLO and PLO by ticking 'V' in the appropriate box.

C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C** = Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Entrepreneurial Skills, **C5** = Ethics & Professionalism

9 Transferable Skills (if applicable)

(Skills learned in the course of study which can be useful and utilized in other settings)

1	Cognitive skills
2	Ethics and Professionalism
3	

Open-ended response (if any)

4	

10 Distribution of Student Learning Time (SLT)

Note: This SLT calculation is designed for home grown programme only.

	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities**									Total SLT
			Face-to-Face (F2F)								NF2F Independent Learning (Asynchronous)	
			Physical				Online/ Technology-mediated (Synchronous)					
			L	T	P	O	L	T	P	O		
1	Chapter 1: Introduction to Artificial Intelligence What is Artificial Intelligence (AI)? Applications of AI AI techniques and approaches	1	4								4	
2	Chapter 2: Knowledge Representation and Reasoning Propositional and first-order logic Semantic networks Frames and scripts Ontologies	1	2		2						5	
3	Chapter 3: Machine Learning Fundamentals Introduction to machine learning supervised, unsupervised, and reinforce learning classification and regression evaluation metric	2	2		2						5	
4	Chapter 4: Neural Networks and Deep Learning Introduction to Neural Networks Perceptrons Multi-layer perceptrons (MLPs) Convolutional Neural Networks (CNNs) Recurrent Neural Networks (RNNs) Deep learning applications	2	4		2						7	
5	Chapter 5: Natural Language Processing (NLP) Introduction to NLP Text preprocessing Language modeling Word embeddings Named Entity Recognition (NER) Sentiment analysis Machine Translation	2	4		2						7	
6	Chapter 6: Planning and Decision Making Markov Decision Processes (MDPs) Dynamic Programming Monte Carlo Methods Reinforcement Learning algorithms (e.g., Q-learning, SARSA)	2	4		2						7	
7	Chapter 7: Computer Vision Introduction to Computer Vision Image processing techniques Feature extraction Object detection and recognition Image segmentation	2	4		2						7	
8	Chapter 8: Ethical and Societal Impact Implications, and Future of AI Bias and fairness in AI AI and employment Advanced AI techniques and research areas	3	4								4	
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
											SUB-TOTAL SLT:	**86**

	Continous Assessement	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment (Asynchronous)
			Physical	Online/ Technology-mediated (Synchronous)	
1	Assignments	20			6
2	Test	20	2		4

3	Project	20			12
4					
5					
					SUB-TOTAL SLT: 24

	Final Assessement	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment (Asynchronous)
			Physical	Online/ Technology-mediated (Synchronous)	
1	Final Examination	40	2.5		7.5
2					
3					
4					
5					
					SUB-TOTAL SLT: 10

	SLT for Assessment:	34
	GRAND TOTAL SLT:	120
A	% SLT for F2F Physical Component: *[Total F2F Physical /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100)]*	37.08
B	% SLT for Online & Independent Learning Component: *[(Total F2F Online + Total Independent Learning) /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]*	**62.92**
C	% SLT for All Practical Component: *[% F2F Physical Practical + % F2F Online Practical]*	**10.00**
C1	% SLT for F2F Physical Practical Component *[Total F2F Physical Practical /(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100)]*	**10.00**
C2	*% SLT for F2F Online Practical Component* *[Total F2F Online Practical / (Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]*	

Please tick (√) if this course is **Industrial Training/ Clinical Placement/ Practicum** using 50% of Effective Learning Time (ELT)

Note:

* Indicate the CLO based on the CLO's numbering in Item 8

** For ODL programme: Courses with mandatory practical requiremnets imposed by the programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

11	Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)	

12	References (include required and further readings, and should be the most current)	- Main References supporting the course: Todd A. Holmes, Cloud Computing 2022 User Guide: A Comprehensive Guide for Beginners, Intermediates, and Experts, Independently published (April 19, 2022), ASIN : B09XZM7MC5, ISBN-13 : 979-8806214387 - Additional References supporting the course:
13	Other additional information (if applicable)	

Note: Number of PLO indicated is purely for illustration purposes only and the number is subjected to the curriculum design.